

$$\frac{\sum (X_i - \bar{X})^2}{\sqrt{Y/n}}$$

Student's t-distribution:

Let X and Y be two independent random variables

$$X \sim N(0,1) \text{ , } Y \sim \chi^2(n)$$

$T = \frac{X}{\sqrt{Y/n}}$ is said to have a student's t-distribution

The pdf of T is

$$f_T(t) = \frac{1}{\sqrt{n} B(\frac{n}{2}, \frac{1}{2})} \left(1 + \frac{t^2}{n}\right)^{-\frac{n+1}{2}}, \quad -\infty < t < \infty.$$

- The density is symmetric about $t=0$. Hence all odd ordered moments vanish (provided they exist).
- Even ordered moments exists of order $< n$.

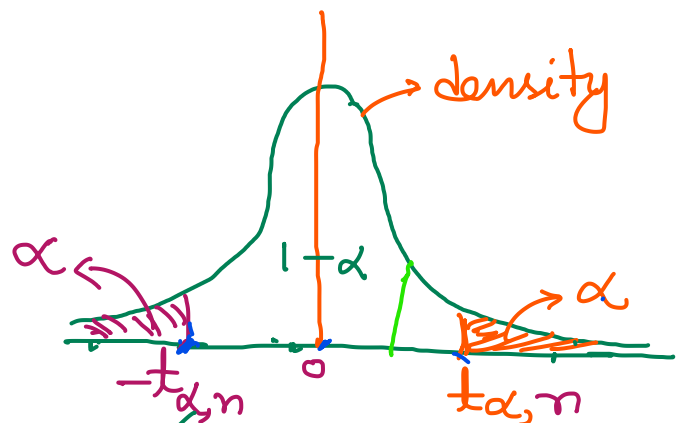
$$E(T) = 0, \therefore E(T^2) = V(T) = \frac{n}{n-2}, \quad n > 2$$

- Measure of kurtosis $\beta_2 = \frac{\mu_4}{\mu_2^2} - 3 = \frac{6}{n-4} > 0$ as $n > 4$

So the density of T is leptokurtic.

$$P(T \geq t_{\alpha, n}) = \alpha$$

upper α 100% point of t_n distribution.



□ Let $X_1, X_2, \dots, X_n \stackrel{i.i.d.}{\sim} N(\mu, \sigma^2)$. Then $\bar{X}$ and S^2 are independent & $\boxed{\bar{X} \sim N(\mu, \frac{\sigma^2}{n})} \Rightarrow$

$$\left. \begin{array}{l} \checkmark \boxed{\sqrt{n} \left(\frac{\bar{X} - \mu}{\sigma} \right) \sim N(0, 1)} \\ \checkmark \boxed{\frac{(n-1) S^2}{\sigma^2} \sim \chi_{n-1}^2} \end{array} \right\} \Rightarrow \frac{\sqrt{n} \left(\frac{\bar{X} - \mu}{\sigma} \right)}{\sqrt{\frac{(n-1) S^2}{\sigma^2} / (n-1)}} \sim t_{n-1}$$

$$\Rightarrow \boxed{\frac{\sqrt{n} (\bar{X} - \mu)}{S} \sim t_{n-1}}$$

$\therefore$ t-distribution is a sampling distribution.

Theorem: Let $T \sim t_n$. As $n \rightarrow \infty$, the pdf of T converges to $\phi(t) = \frac{1}{\sqrt{2\pi}} e^{-\frac{t^2}{2}}$, $-\infty < t < \infty$

The F-distribution: Let Y_1 and Y_2 be independently distributed. & $Y_1 \sim \chi_m^2$, $Y_2 \sim \chi_n^2$. Then $U = \frac{Y_1/m}{Y_2/n} = \frac{n}{m} \frac{Y_1}{Y_2}$ is said to have a F-distribution on (m, n) d.f.

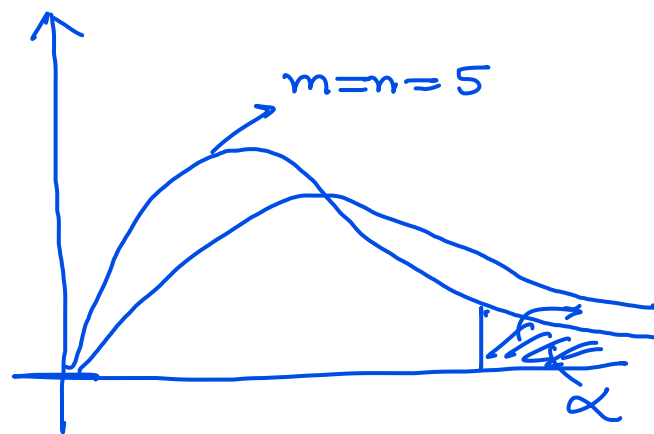
The pdf of U is

$$f_U(u) = \left(\frac{m}{n} \right)^{\frac{m}{2}} \frac{1}{B\left(\frac{m}{2}, \frac{n}{2}\right)} \frac{u^{\frac{m}{2} - 1}}{\left(1 + \frac{m}{n} u\right)^{\frac{m+n}{2}}}, u > 0$$

$$E(U) = \frac{n}{n-2}, \quad n > 2$$

$$V(U) = \frac{2n^2(m+n-2)}{m(n-2)^2(n-4)}, \quad n > 4$$

positively skewed distribution



I. To test the hypothesis: $H_0: \mu = \mu_0$
vs $H_1: \mu \neq \mu_0$

$x_1, x_2, \dots, x_{20}$

II. Under the null hypothesis H_0 , the test statistic

$$T = \frac{\sqrt{n}(\bar{x} - \mu_0)}{s} \text{ needs to be computed.}$$

$$= \frac{\sqrt{20}(\bar{x} - 60)}{2} = \bar{x}$$

III. Reject H_0 if $|T| > t_{\frac{\alpha}{2}, n-1}$

$$\underline{T > t_{\frac{\alpha}{2}, n-1} \text{ or } T < -t_{\frac{\alpha}{2}, n-1}}$$

$X_1, X_2, \dots, X_m \stackrel{\text{i.i.d.}}{\sim} N(\mu_1, \sigma_1^2)$
 $Y_1, Y_2, \dots, Y_n \stackrel{\text{i.i.d.}}{\sim} N(\mu_2, \sigma_2^2)$ > independent

$$U = \frac{(m-1)S_X^2}{\sigma_1^2} \sim \chi_{m-1}^2,$$

$$V = \frac{(n-1)S_Y^2}{\sigma_2^2} \sim \chi_{n-1}^2$$

$$\therefore \frac{U/(m-1)}{V/(n-1)} \sim F_{m-1, n-1}$$

$$\Rightarrow \boxed{\frac{\sigma_2^2 S_X^2}{\sigma_1^2 S_Y^2} \sim F_{m-1, n-1}}$$

$$H_0: \sigma_1^2 = \sigma_2^2$$

$$H_0: \sigma_1^2 > \sigma_2^2$$